

The empirical $K_{\text{ASP}}(N)$ dictionary: a numerically verified correspondence between class-number- N imaginary quadratic fields and $\text{SU}(N)$ centre data

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Abstract

For each integer $N \geq 1$ we define $K_{\text{ASP}}(N) = \mathbb{Q}(\sqrt{D_{\min}^{(N)}})$ where $D_{\min}^{(N)}$ is the smallest absolute value of a fundamental negative discriminant D such that $\mathbb{Q}(\sqrt{D})$ has class number $h_K = N$ and 2-rank $\text{rk}_2(\text{Cl}(K)) = v_2(N)$. We tabulate $K_{\text{ASP}}(N)$ via PARI/GP for $N \in \{1, \dots, 30\} \cup \{36, 40, 44, 48\}$, and we report $N = 32$ as an open numerical case (no admissible $|D| \leq 2 \times 10^6$ found). Cohen-Lenstra-Gerth-Smith heuristics provide a motivation, not a proof, for the existence of $K_{\text{ASP}}(N)$ at large N . The dictionary exhibits a structural compatibility with the centre $Z(\text{SU}(N)) = \mathbb{Z}/N\mathbb{Z}$: $\text{Cl}(K_{\text{ASP}}(N))$ matches the cyclic and 2-Sylow structure of Z_N . This compatibility is informal and does not constitute a theorem of arithmetic-physics correspondence.

1 Introduction

The Cohen-Lenstra heuristics [1] predict statistical distributions of class groups of imaginary quadratic fields. In particular, for fixed odd primes p , the p -rank of $\text{Cl}(K)$ follows a specific limiting distribution. The case $p = 2$ is special, controlled by Gauss's genus theory [2]: if D is a fundamental discriminant with $t = \omega(D)$ distinct prime divisors, then

$$\text{rk}_2(\text{Cl}(K)) = t - 1.$$

This paper investigates a specific selection rule connecting integer indices N to imaginary quadratic fields via simultaneous constraints on class number and 2-rank.

2 Main Theorem

Definition 1. For integer $N \geq 1$, set $v_2(N)$ to be the largest power of 2 dividing N . Define

$$K_{\text{ASP}}(N) = \min\{|D| : D < 0, D \text{ fund. disc.}, h_K = N, \text{rk}_2(\text{Cl}(K)) = v_2(N)\}$$

where $K = \mathbb{Q}(\sqrt{D})$ and the minimum is taken over absolute values. We say $K_{\text{ASP}}(N)$ is defined if the set is non-empty.

Proposition 1 (Tautological well-definedness). *Whenever the admissibility set $\{D < 0 \text{ fund. disc.} : h_K = N, \text{rk}_2(\text{Cl}(K)) = v_2(N)\}$ is non-empty, the minimum $|D_{\min}^{(N)}|$ exists by well-ordering of $\mathbb{N}$, and $K_{\text{ASP}}(N)$ is well defined.*

Remark 1 (Existence: empirical status). We have verified by PARI/GP that the admissibility set is non-empty for $N \in \{1, \dots, 30\} \cup \{36, 40, 44, 48\}$. The case $N = 32$ is open: no admissible discriminant with $|D| \leq 2 \times 10^6$ has been found. Cohen–Lenstra heuristics [1] motivate, but do not prove, non-emptiness at large N . We do not claim a general existence theorem.

Table 1: $K_{\text{ASP}}(N)$ for $N \in \{1, \dots, 30\} \cup \{32, 36, 40, 44, 48\}$ (PARI verified, $N = 32$ open)

N	$v_2(N)$	D	$\text{Cl}(K)$	N	$v_2(N)$	D	$\text{Cl}(K)$
1	0	−3	trivial	16	4	−5460	$(\mathbb{Z}/2)^4$
2	1	−15	$\mathbb{Z}/2$	17	0	−383	$\mathbb{Z}/17$
3	0	−23	$\mathbb{Z}/3$	18	1	−335	$\mathbb{Z}/18$
4	2	−84	$(\mathbb{Z}/2)^2$	19	0	−311	$\mathbb{Z}/19$
5	0	−47	$\mathbb{Z}/5$	20	2	−455	$\mathbb{Z}/10 \times \mathbb{Z}/2$
6	1	−87	$\mathbb{Z}/6$	24	3	−2184	$\mathbb{Z}/6 \times (\mathbb{Z}/2)^2$
7	0	−71	$\mathbb{Z}/7$	28	2	−935	$\mathbb{Z}/14 \times \mathbb{Z}/2$
8	3	−420	$(\mathbb{Z}/2)^3$	32	5	open ($ D > 2 \times 10^6$)	—
9	0	−199	$\mathbb{Z}/9$	36	2	−1295	$\mathbb{Z}/18 \times \mathbb{Z}/2$
10	1	−119	$\mathbb{Z}/10$	40	3	−2415	$\mathbb{Z}/10 \times (\mathbb{Z}/2)^2$
11	0	−167	$\mathbb{Z}/11$	44	2	−2135	$\mathbb{Z}/22 \times \mathbb{Z}/2$
12	2	−231	$\mathbb{Z}/6 \times \mathbb{Z}/2$	48	4	−11220	$\mathbb{Z}/6 \times (\mathbb{Z}/2)^3$

3 Connection to Cohen-Lenstra Heuristics

Conjecture 1 (Asymptotic upper bound). *For $N = 2^k$ a pure power of 2, $|K_{\text{ASP}}(N)| \leq \prod_{p \leq P_k} p$ where P_k is the k -th prime. In particular,*

$$|K_{\text{ASP}}(16)| \leq 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310, \text{ observed } 5460,$$

$$|K_{\text{ASP}}(32)| \leq 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 30030.$$

The observed values exceed the naive primorial bound by a factor depending on Cohen-Lenstra density.

4 Arithmetic-Physics Correspondence (Informal)

The dictionary K_{ASP} exhibits a structural compatibility with the centre Z_N of the gauge group $\text{SU}(N)$:

- For N odd, $\text{Cl}(K_{\text{ASP}}(N))$ is cyclic of order N , mirroring the cyclic centre $Z_N \cong \mathbb{Z}/N\mathbb{Z}$.
- For $N = 2^k$, $\text{Cl}(K_{\text{ASP}}(N))[2] = (\mathbb{Z}/2)^k$, mirroring the 2-Sylow subgroup of Z_N .
- For mixed $N = 2^k \cdot m$ (m odd), the dictionary selects the minimal field whose class group contains both cyclic and 2-torsion components matching Z_N .

This correspondence suggests $K_{\text{ASP}}(N)$ as the natural arithmetic object associated to $\text{SU}(N)$ in the context of TQFT-Arakelov frameworks.

5 Volumes of $X_N = H^3/PSL_2(O_{K_{\text{ASP}}(N)})$

Using Humbert's formula [3]:

$$\text{Vol}(X_N) = \frac{|d_K|^{3/2} \cdot \zeta_K(2)}{4\pi^2}$$

we compute (PARI verified):

N	D	$\zeta_K(2)$	$\text{Vol}(X_N)$
1	-3	1.285	0.169
2	-15	2.133	3.139
3	-23	2.308	6.449
4	-84	1.730	33.728
8	-420	1.667	363.368
16	-5460	1.614	16491.04

6 Application: an empirical scaling relation

We define the arithmetic surrogate scaling

$$\frac{m_{\text{arith}}(N)}{\sqrt{\sigma_0}} := C(N) \cdot \sqrt{\lambda_1(X_N)}, \quad C(N) = \sqrt{2\pi e \cdot 2/3 \cdot (9/10) \cdot (N^2+1)/N^2},$$

where $X_N = \mathbb{H}^3/\text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)})$ is the Bianchi 3-orbifold and $\lambda_1(X_N)$ is the first non-zero eigenvalue of the hyperbolic Laplacian. This is a *definition*, not a theorem.

Remark 2 (Empirical match, no universal Selberg-type bound invoked). The companion paper [7] reports a root-mean-square agreement of 0.85% between $m_{\text{arith}}(N)/\sqrt{\sigma_0}$ (taking $\lambda_1 = 1$ as a working assumption) and the AT 2021 continuum-extrapolated lattice $m_{0++}/\sqrt{\sigma}$ over six anchors $N \in \{2, 3, 4, 5, 6, \infty\}$ [6]. We make no claim of a universal positive lower bound on $\lambda_1(X_N)$ over all N ; Sarnak’s 1983 paper [5] establishes the Selberg-type bound $\lambda_1 \geq 21/25$ for *specific* arithmetic 3-manifolds, and its applicability to every Bianchi orbifold in the anchor list is an open question. Kim 2003 [4] would conditionally improve this bound on the cuspidal forms in the image of base change from $\text{GL}_2/\mathbb{Q}$. The empirical match is presented as a numerical observation, not as a theorem of physical identification.

7 Open Problems

1. $K_{\text{ASP}}(32)$. Verify whether a fundamental discriminant $|D| \leq 10^9$ satisfies $h_K = 32$ and $\text{rk}_2(\text{Cl}(K)) = 5$, or prove non-existence at all scales. Extensive PARI scan up to $|D| < 2 \times 10^6$ has not produced a candidate.
2. $K_{\text{ASP}}(N)$ **general non-emptiness**. Prove (or disprove) that the admissibility set is non-empty for every N compatible with Gauss genus theory. Cohen–Lenstra–Gerth–Smith heuristics motivate, but do not prove, non-emptiness at large N .
3. **Selberg-type bound**. Does any universal positive lower bound $\lambda_1(X_N) \geq c > 0$ hold uniformly for every Bianchi orbifold in the anchor list, with arbitrary class number $h_K = N$? Sarnak 1983 [5] establishes such a bound for specific arithmetic 3-manifolds. Kim 2003 [4] would conditionally improve the bound via base change.
4. **Empirical scaling note**. Whether the numerical match between $m_{\text{arith}}(N)/\sqrt{\sigma_0}$ (under $\lambda_1 = 1$) and the AT 2021 lattice $m_{0++}/\sqrt{\sigma}$

(RMS 0.85% on six anchors $N \in \{2, \dots, 6, \infty\}$ [6, 7]) admits a structural derivation from 4D pure $SU(N)$ Yang–Mills is open.

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References

- [1] H. Cohen, H.W. Lenstra, *Heuristics on class groups of number fields*, Lecture Notes in Math. **1068**, Springer, 1984.
- [2] C.F. Gauss, *Disquisitiones Arithmeticae*, 1801.
- [3] G. Humbert, *Sur la mesure des classes d’Hermite*, Comptes Rendus Acad. Sci. Paris (1919).
- [4] H. Kim, *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , J. Amer. Math. Soc. **16** (2003), 139–183.
- [5] P. Sarnak, *The arithmetic and geometry of some hyperbolic three-manifolds*, Acta Math. **151** (1983), 253–295. (Scope: specific arithmetic 3-manifolds.)
- [6] A. Athenodorou, M. Teper, *The glueball spectrum of $SU(N)$ gauge theories in $3+1$ dimensions*, JHEP **11** (2021) 172; arXiv:2106.00364 [hep-lat].
- [7] K. Rémondière, *An empirical scaling relation between $SU(N)$ lattice glueball masses and class-number- N imaginary quadratic Bianchi spectra*, companion paper, 2026.
- [8] H.M. Stark, *On complex quadratic fields with class number two*, Math. Comp. **29** (1975), 289–302.
- [9] C. Maclachlan, A.W. Reid, *The Arithmetic of Hyperbolic 3-Manifolds*, Graduate Texts Math. **219**, Springer, 2003.